who do you feel confident you could explain the internet to?

How it works

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Q1

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CS 340

Network Introduction

Updates

1. MP7 ~~Wallet~~ due Next Tuesday

1. HW 6 due next Wednesday

1. Final project check-off May 11th at 1:30pm 5%

MP 10

2-3 hours

Networks

Today’s LGs - Fill in the gaps in your understanding of how the internet works

Be able to answer big picture questions about the internet

Agenda

1. What is a System
2. The internet
    a. Ports
    b. Client-Server
    c. IP addresses
    d. Packet format
    e. IP
    f. UPD
    g. TCP

How would you define a System?

A CPU? ✓

A transistor? ✗

A computer? ✓

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So Far

Individual computer system - CPU, RAM, Disk

virtual memory

schedulings

caching

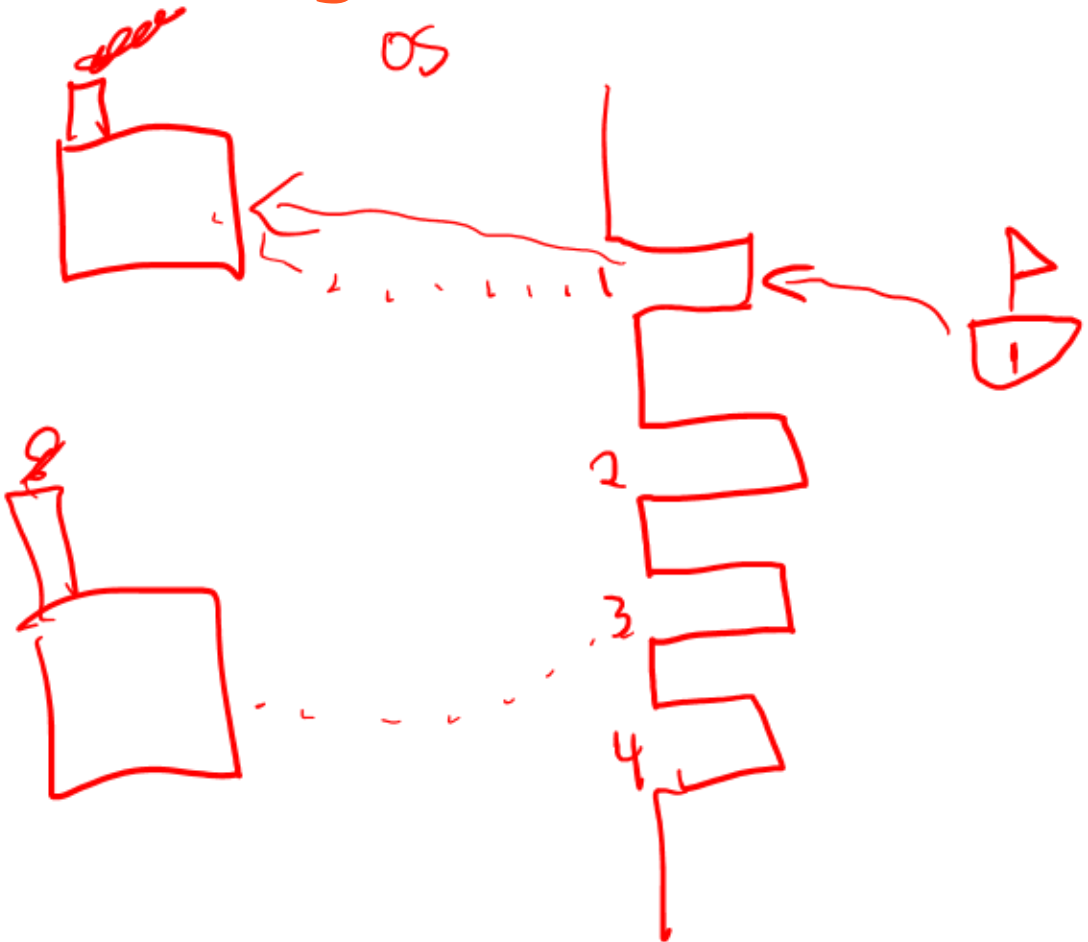
The OS kernel code manges...

next → OS manages how a process talks to another process → on a dif computer

The OS manages incoming information through...

Ports

- number
- address



Some ports are...

443 – listening for HTTPS request

Ports 1024–49151 applications to use

49152+ ephemeral ports

OS can use

Who manages the port system on your computer?

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A) The user
B) External servers
C) The OS
D) The internet

Client-Server Design

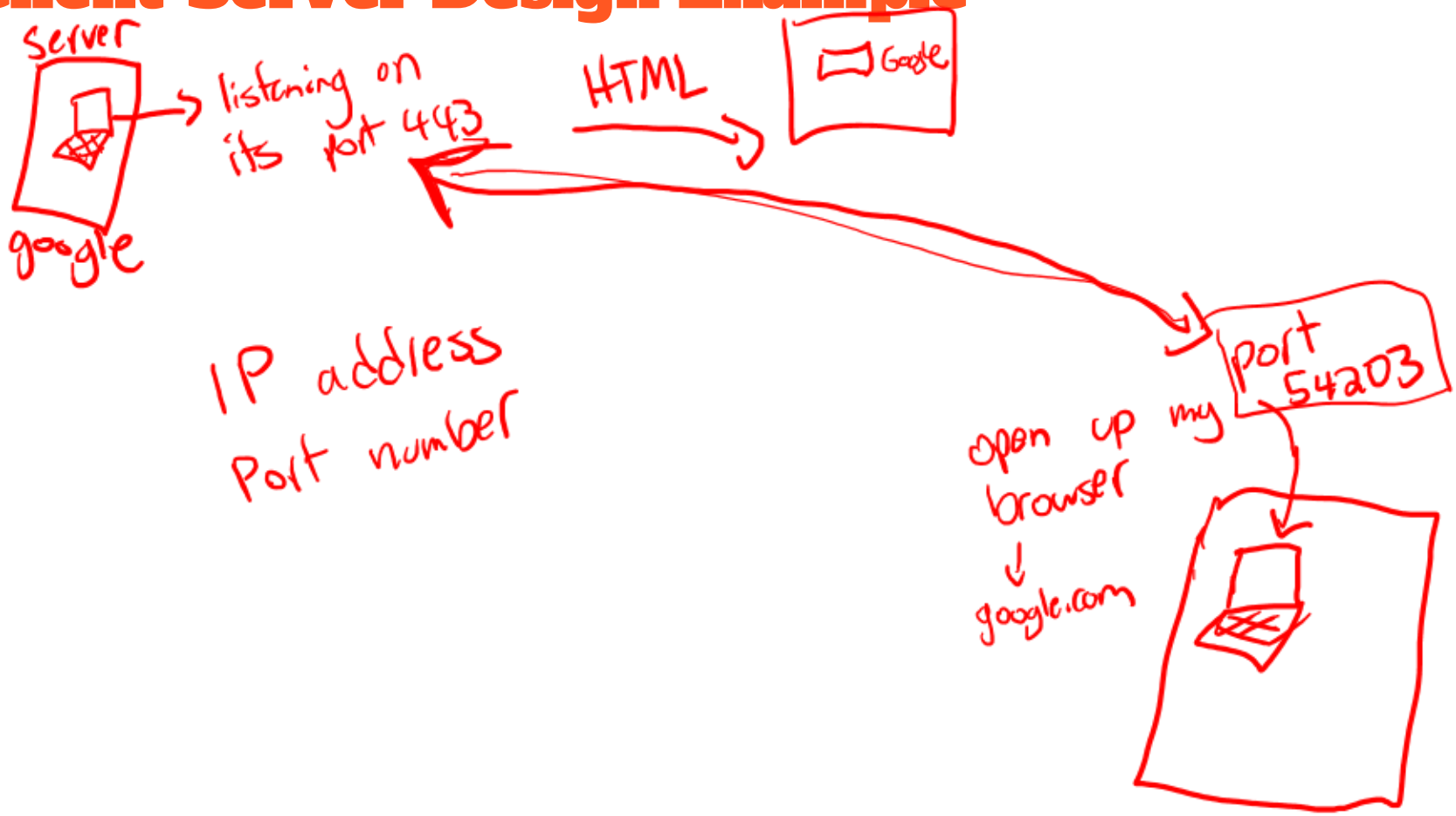
server – a process that listens for incoming requests at a port

client – a process thats initiates contact

Server receives request then establishes a connection between server port and clients port

– need to communicate in both directions

Client-Server Design Example



You open up two different websites in your browser. How does your OS know which incoming packets belong to which tab?

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A) Each tab uses a different well-known port
B) Each connection has a unique ephemeral port on your machine
C) The middle byte of an intThe server labels each packet with the tab number
D) Your browser assigns each tab a different IP address

Internet

The internet is... a physical infrastructure

specialized computers

connected system of computers and routers

that move bytes via wires and waves

Let's say a router...

Gets the bytes:

0x504DF45A1BB3194BE504DF45A1BB3194BE504DF45A1BB3194BE504DF45A1BB3194BE504DF45A1BB3194BE504DF45A1BB3194BE504DF45A1BB3194BE504DF45A1BB3194BE504DF45A1BB3194BE504DF45A1BB3194BE504DF45A1BB3194BE504DF45A1BB3194BE504DF45A1BB3194BE504DF45A1BB3194BE504DF45A1BB3194BE

We need a protocol ← agreed upon format

where should they go

what is the data ...

where does one packet start and end.

What do you think is being sent through wires and radio signals from router to router?

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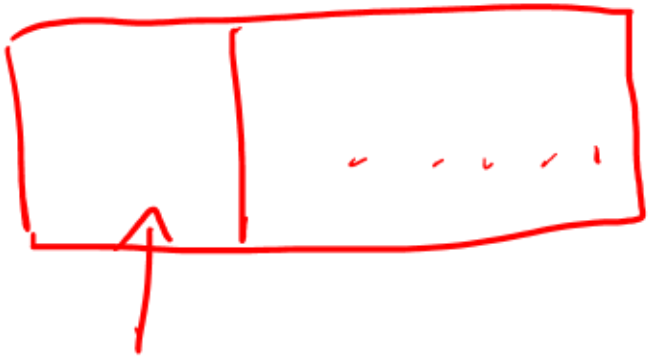
A) 2 signals, either 1 or 0
B) 16 different signals to make up a hex value
C) Voice audio saying a hex value
D) Morse Code

Internet Protocol (IP)

Packets – small

Packet

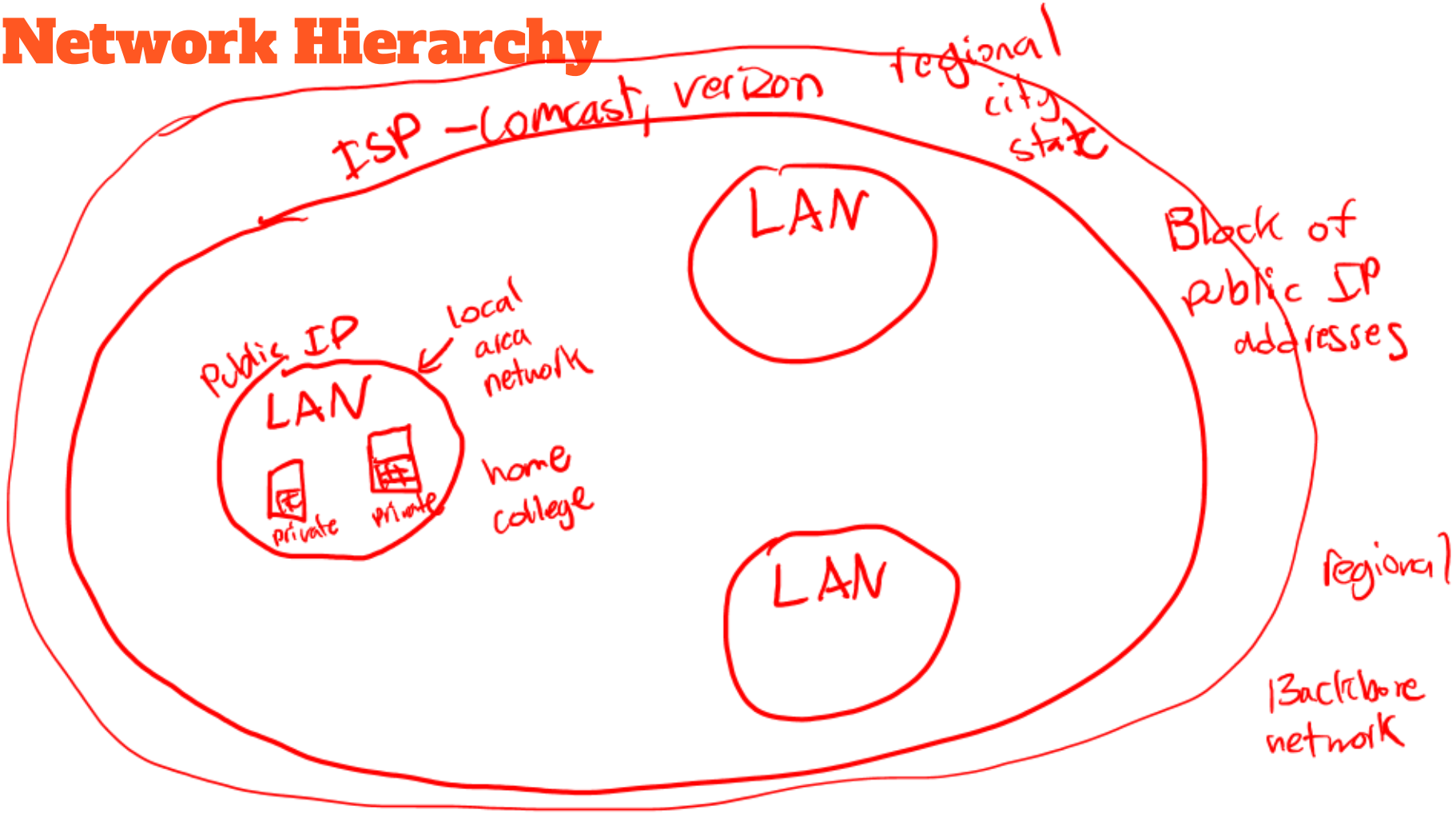


IP Header

Global IP

– source and destination addresses

Network Hierarchy



How many IP addresses do you think we currently need? Roughly around...

population is 8.2 billion

public

~3.7 billion IP addresses

A) 100k
B) 1 million
C) 1 billion
D) 1 trillion

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IP Addresses

IPv4 → 32 bit addresses

1 or 0 _ _ _ _ _ _ _ ... 32

→ 4.3 billions

IPv6 – 128 bit addresses

IPv4

172.16.0.1

IPV6

2001:0db8:85a3:0000:0000:8a2e:0370:7334

If IPv4 with 32 bit addresses can represent ~4 ~~million~~ billion addresses, how many can IPv6 with a 128 bit address represent?

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A) (128/32) * 4 million aka 16 million
B) (128 - 32) * 4 million aka 384 million
C) 2^128 aka 3.4028237e+38
D) 1 trillion

DNS (domain name system)

type google.com

- check a cache
- if not asks a DNS server
  - DNS server look up the IP address for google

domain name → IP address

```
drschatz@cs-drschatz-MBP ~ % nslookup google.com
Server:         2001:558:feed::1
Address:        2001:558:feed::1#53

Non-authoritative answer:
Name:   google.com
Address: 142.250.217.110

drschatz@cs-drschatz-MBP ~ %
```

IPv6 → 2001:558:feed::1 ← DNS server IP

port

IPv4

port #443

IPv6 Packet

40 byte IP header

20 byte TCP header

→ Payload – 1.4 KB (max)

dest IP
source IP

used by the routers

Why do you think there is a size limit to packets?

- error recovery – if it needs to be resent it's small
- fairness
- physical constraints – hardware buffers

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Layers of a packet

payload

Layer	Protocols	Use
Application	HTTP, SSH, SMTP, etc.	programmer-visible messages
Transport	**TCP, UDP**	reach a specific program
Internet	**IP**, which has two versions: IPv4 and IPv6	reach a specific computer

UDP - User Datagram Protocol

- always have IP layer above

- small
- dest port

- Not guaranteed to arrive and in any order
- fast

Video calling

TCP - Transmission Control Protocol

Reliable and ordered

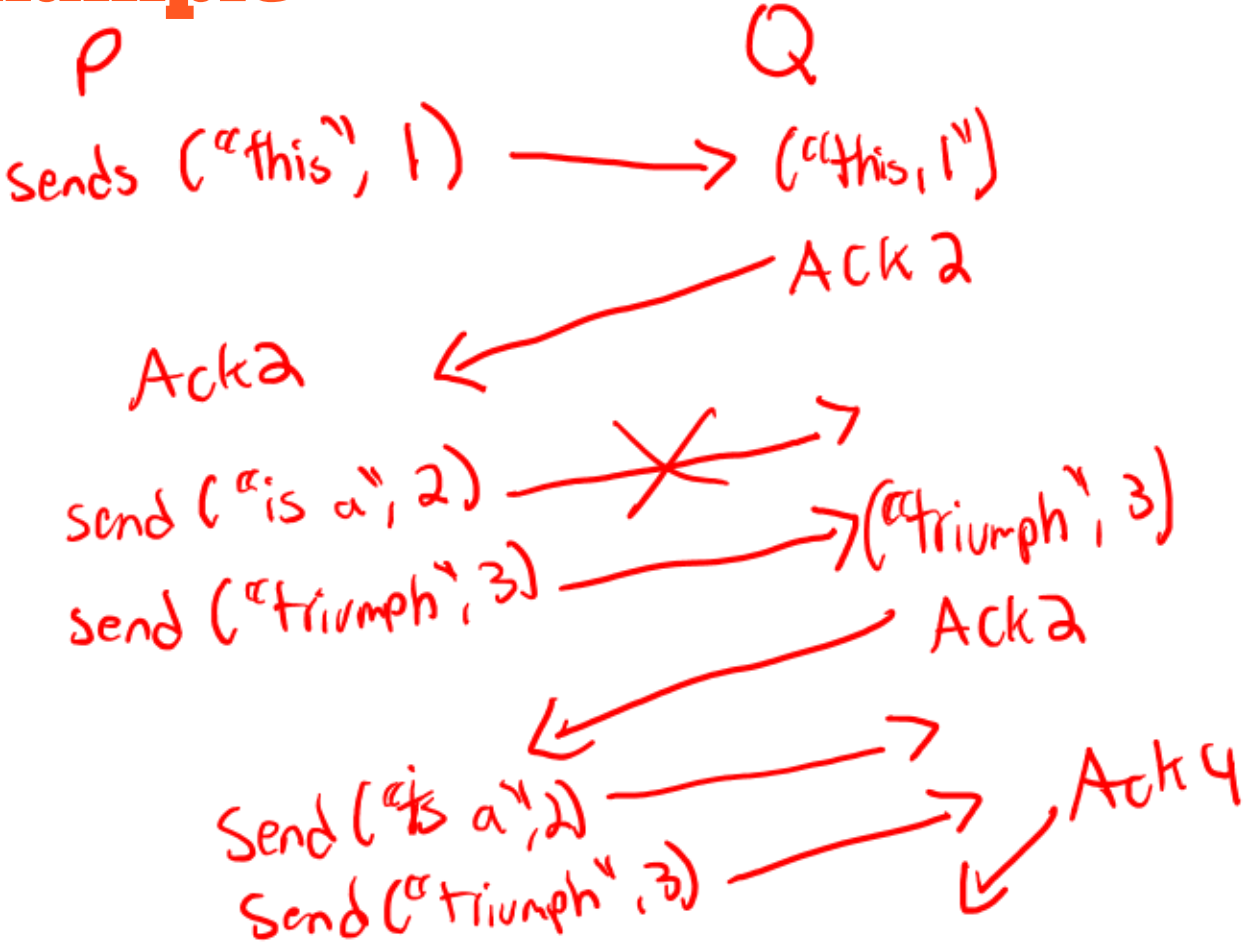
server-client handshake

client → "hey here are my sequence start number" → 10

server → "I see that, I'm starting at 512"

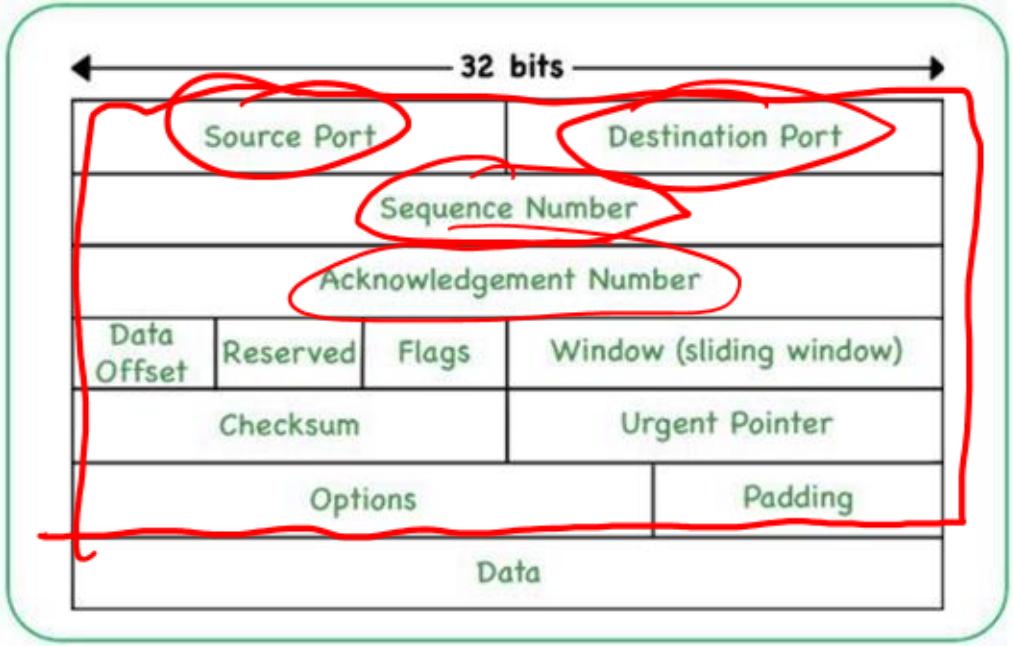
client → "I see you"

TCP Example

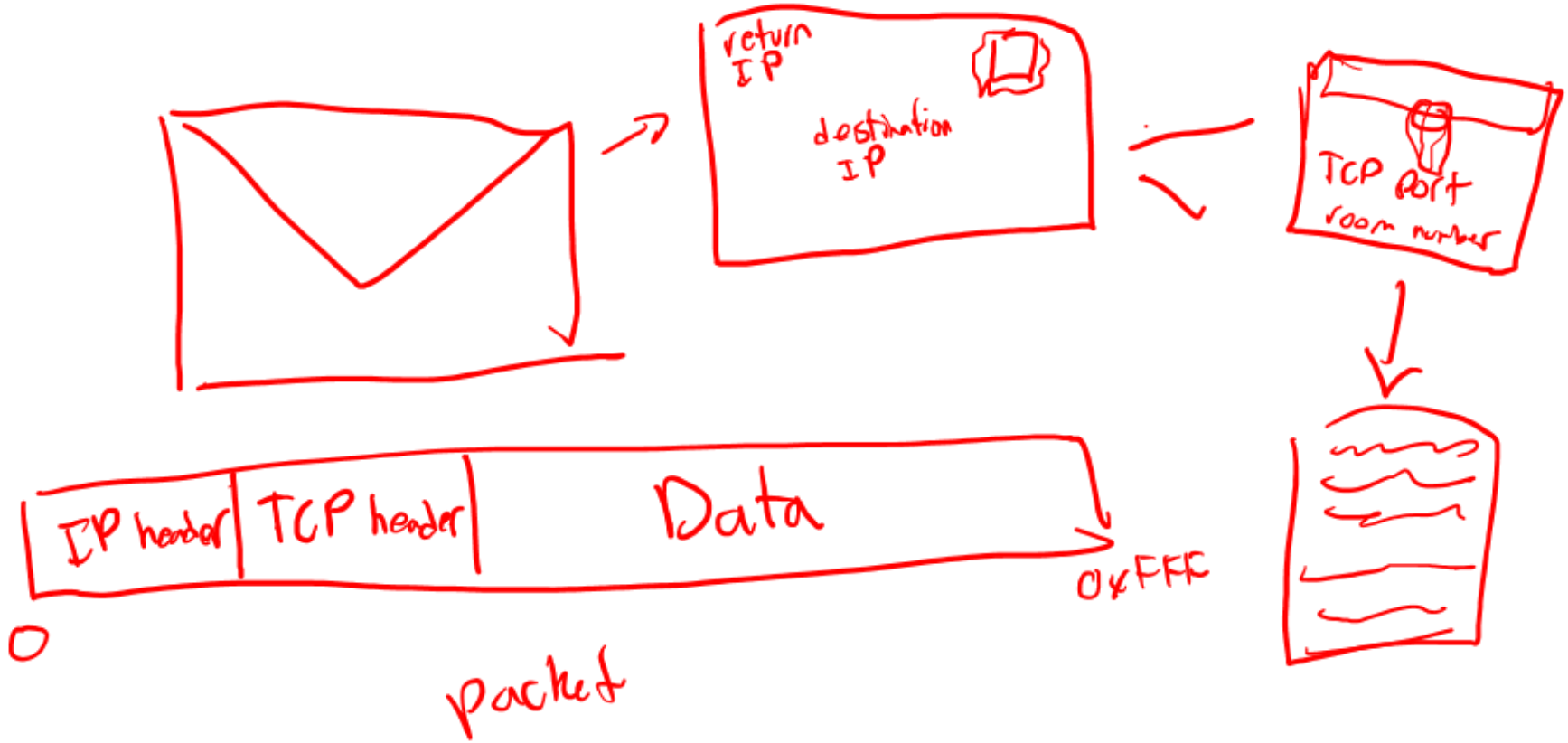

P
Q
sends ("this", 1)
("this, 1")
ACK 2
Ack2
send ("is a", 2)
("triumph", 3)
send ("triumph", 3)
Ack2
Send ("is a", 2)
Ack 4
Send ("triumph", 3)

TCP Header Format

IP header, TCP header, data...



Packet Overview



Which do you think is faster for getting data from a server to your computer?

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A) IP + UDP

B) IP + TCP

~~C) Only TCP~~

~~D) Only UDP~~

Which are True?

A) Packets are just bytes T
B) I need to use the IP protocol to send data over the internet. T
C) I need to use the TCP protocol to send data over the internet F
D) My OS uses ~~IP addresses~~ Ports to figure out where to send incoming packets internally
E) My LAN has a private IP address only I know. F
F) The internet is a system of computers T

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Q10

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